

CRED-nf checklist summary

18 July, 2026

Manuscript title: Hybrid versus Intermittent Feedback in Alpha Neurofeedback Training: A Single-Session Regulation and Discrimination Study

Corresponding Author: Pablo Alexis Olguin Baxman, Wolf Pink, Prof. Dr. Stefan Debener

Corresponding author email: alexisbaxman@gmail.com, wolf.pink@uni-oldenburg.de, stefan.debener@uni-oldenburg.de

Item No.	Checklist item	Manuscript Details
Pre-experiment		
1a	Pre-register experimental protocol and planned analyses	Methods: “The study was preregistered on the Open Science Framework (OSF): https://osf.io/2qg5j .”
1b	Justify sample size	Methods: “A total of 33 participants were recruited and recorded at the Neuropsychology Laboratory of the Carl von Ossietzky University of Oldenburg and they were compensated with 12€ per hour for their participation. (...) Sample size was determined a priori: a power analysis indicated that 27 participants would be required to achieve 80% power for detection of a difference between conditions. We anticipated a 10% exclusion rate, therefore recruitment targeted a sample size above 30. Six participants were excluded from subsequent analyses due to insufficient alpha reactivity, defined as the absence of the expected alpha amplitude increase from eyes-open to eyes-closed during the baseline (see Supplementary Material A). The final sample consisted of 27 participants (10 male and 17 female) with ages ranging from 19 to 49 years old ($M = 27.37$, $SD = 5.53$). Two participants were left-handed. No participants reported taking medication or having a history of neurological or psychiatric disorders.”
2a	Employ control group(s) or control condition(s)	Methods: “The study used a within-subject design with two feedback conditions: hybrid (continuous sonification plus intermittent reward tones) and intermittent (reward tones only).” The intermittent condition served as a control condition, isolating the effect of continuous sonification while holding the intermittent reward component constant across conditions. As noted in the discussion, there is a no-feedback control condition, which cannot rule out effects attributable to the intermittent feedback in general. Blinding was not applicable to condition assignment, as participants could directly perceive the presence or absence of continuous auditory feedback.
2b	When leveraging experimental designs where a double-blind is possible, use a double-blind	<i>NA: A double-blind was not appropriate for this experiment</i>
2c	Blind those who rate the outcomes	<i>NA: There was only one participant group</i>

	Blind those who analyse the data	<i>Those who analysed the data were not blind to group assignment</i>
2d	Examine to what extent participants and experimenters remain blinded	<i>NA: There was only one participant group</i>
2e	In clinical efficacy studies, employ a standard-of-care intervention group as a benchmark for improvement	<i>NA: This is not a clinical efficacy study</i>

Control measures

3a	Collect data on psychosocial factors	<i>Psychosocial factors were not measured</i>
3b	Report whether participants were provided with a strategy	Methods: “Written task instructions were displayed at the start of the session and repeated before every task (see Supplementary Material B).” Supplementary Material B: “Your goal is to manipulate your alpha waves. Initially, you will focus on increasing your alpha state by relaxing and maintaining a calm mind... After a while, you will hear the voice instructing you to decrease your alpha state. You can try this by thinking about something stressful, remembering a tense moment, or imagining physical activity.”
3c	Report the strategies participants used	<i>The strategies participants used were not recorded or not reported in the manuscript</i>
3d	Report methods used for online-data processing and artifact correction	Methods: “The ongoing absolute alpha amplitude (AAA) was estimated using a ring buffer structure, creating a 4-second sliding window with 90% overlap, updated at a sampling rate of 2.5 Hz. On each update, the entire buffer was filtered between 8-12 Hz using a bandpass-filter, then the instantaneous amplitude envelope was calculated and averaged to obtain an estimate of the AAA during the last 4 seconds. The online pipeline introduced an estimated total delay of approximately 3.08 seconds, and it had an update rate of 2.448 Hz (SD = 0.0009 Hz), 1 sample every 0.4 seconds. This estimate was then used for the alpha-state evaluation and the feedback calculation. For more details on the ring buffer and the online analysis, see Supplementary Material C.”
3e	Report condition and group effects for artifacts	<i>Condition and group effects for artifacts were not measured, or not reported in the manuscript</i>

Feedback specifications

4a	Report how the online-feature extraction was defined	Methods: “Both the window duration and the use of absolute rather than relative amplitude were chosen following the recommendation by Frederick (2012) to improve discrimination accuracy. The buffer contained data exclusively from the O2 electrode, as alpha activity is typically stronger at occipital sites (Frederick, 2012; Nunez, 2005).”
----	--	---

4b	Report and justify the reinforcement schedule	Methods: “Both tasks under both conditions included intermittent feedback in the form of a reward sound contingent on successful trials. The reward sound chosen was the ”Tada.wav” (Microsoft Corporation, 1992) as in Frederick et al. (2016).” “A [regulation] trial was considered successful when the participant’s alpha amplitude remained above (up-regulation) or below (down-regulation) a dynamic threshold for a period of 4 seconds, triggering the delivery of a reward sound.” In the discrimination task: “Participants then indicated whether they perceived their alpha state as high or low. Correct responses were followed by the delivery of the same reward sound as in the regulation task.”
4c	Report the feedback modality and content	Methods: “Both tasks under both conditions included intermittent feedback in the form of a reward sound contingent on successful trials. The reward sound chosen was ”Tada.wav” (Microsoft Corporation, 1992), as in Frederick et al. (2016). In the hybrid condition, the regulation task additionally included continuous auditory feedback: a tone whose volume tracked ongoing alpha amplitude in real time.” “The continuous auditory feedback was implemented by mapping the ongoing AAA onto the volume of a continuous bandpass-filtered red noise tone, generated using Purr-Data (Gräf, 2024). A direct mapping approach was chosen to keep instructions intuitive: higher alpha amplitude corresponding to higher sound volume. Prior to mapping, the AAA was min-max normalized to the range [0, 1] within an interval defined symmetrically around the dynamic threshold and clipped at the lower or upper edges.”
4d	Collect and report all brain activity variable(s) and/or contrasts used for feedback, as displayed to experimental participants	Methods: ”This estimate [Absolute Alpha Amplitude (AAA)] was then used for the alpha-state evaluation and the feedback calculation [continuous auditory feedback].” ”The continuous auditory feedback was implemented by mapping the ongoing AAA onto the volume of a continuous bandpass-filtered red noise tone, generated using Purr-Data (Gräf, 2024).”
4e	Report the hardware and software used	Methods: “The experiment was programmed in MATLAB R2022b (The MathWorks, Inc., 2022a) using the Psychtoolbox v3.0.18 extension (Kleiner et al., 2007), EEGLAB 2025.0.0 (Delorme & Makeig, 2004) and Lab Streaming Layer (LSL) (Kothe et al., 2025). Participants’ EEG were recorded using a SMARTING Pro mobile EEG system (mBrainTrain, 2025) with 32 electrodes arranged according to the international 10-20 system.”
Outcome measures - brain		
5a	Report neurofeedback regulation success based on the feedback signal	Methods: “For the regulation task, the criterion was defined as a at least a 10% alpha amplitude difference between the AAA during the up- and during the down-regulation trials (see Supplementary Material H).”

5b	Plot within-session and between-session regulation blocks of feedback variable(s), as well as pre-to-post resting baselines or contrasts	Supplementary Material I: “Figure S8: Normalized absolute alpha amplitude (AAA) across the session. The yellow violin plot shows baseline AAA vales and the four dark cyan violin plot show AAA during Block 1-4. Gray lines connect individual participants’ values across blocks, pink lines indicate participants whose block-level AAA exceeded their own baseline value in that specific block.”
5c	Statistically compare the experimental condition/group to the control condition(s)/group(s) (not only each group to baseline measures)	Results: “A paired t-test indicated no significant difference in amplitude difference between the hybrid and intermittent conditions, $t(26) = 1.039$, $p = .851$, Cohen’s $d = 0.200$. “
Outcome measures - behaviour		
6a	Include measures of clinical or behavioural significance, defined a priori, and describe whether they were reached	<i>The manuscript does not include measures of clinical or behavioural significance</i>
6b	Run correlational analyses between regulation success and behavioural outcomes	Results: “On average, performance in the regulation task was slightly higher in the hybrid condition ($M = 60.419\%$, $SD = 11.401$) than in the intermittent condition ($M = 58.914\%$, $SD = 12.186$), see Figure 4a. However, the paired t-test indicated no significant effect of the hybrid feedback relative to only the intermittent feedback, $t(26) = 0.551$, $p = 0.585$, Cohen’s $d = 0.106$. Similarly, balanced accuracy in the discrimination task was slightly higher in the hybrid condition ($M = 49.395\%$, $SD = 13.211$) than in the intermittent condition ($M = 47.618\%$, $SD = 12.773$), see Figure 4b. However, the paired t-test indicated no significant effect of prior exposure to the hybrid feedback relative to only the intermittent feedback, $t(26) = 0.548$, $p = 0.588$, Cohen’s $d = 0.105$.”
Data storage		
7a	Upload all materials, analysis scripts, code, and raw data used for analyses, as well as final values, to an open access data repository, when feasible	Methods: “Experiment code, analysis scripts, and study materials are available at github.com/alexisbaxman/alpha-neurofeedback-hybrid-intermittent . Raw participant data are not publicly available, as participant consent did not cover public data sharing.”